

ISLAMABAD INTL, Pakistan

WMO: 415710

Lat: 33.617N

Lon: 73.099E

Elev: 1668

StdP: 13.83

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.2	38.4	26.4	21.6	61.5	29.8	25.4	54.1	26.3	57.5	23.0	59.7	0.1	230	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	23.3	105.9	73.1	102.6	73.4	100.4	73.0	82.5	93.8	81.5	92.5	80.5	91.1	13.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.7	164.2	88.6	78.9	159.5	88.1	77.6	153.0	87.0	48.2	93.4	47.0	93.2	45.8	91.3	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.0	23.2	20.7	32.7	110.6	1.7	3.1	31.5	112.8	30.6	114.7	29.6	116.4	28.4	118.7
			31.4	85.2	1.9	1.7	30.1	86.4	28.9	87.4	27.9	88.4	26.5	89.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.3	51.9	56.8	66.1	75.1	84.2	88.6	86.1	84.2	81.7	74.2	63.0	54.7
	DBStd	13.51	3.67	4.65	5.78	6.20	5.69	4.79	4.43	3.53	3.63	4.87	4.14	3.63
	HDD50	28	21	3	0	0	0	0	0	0	0	0	0	4
	HDD65	1103	405	231	56	5	0	0	0	0	0	1	86	319
	CDD50	8170	80	194	499	754	1062	1157	1121	1060	952	750	390	151
	CDD65	3771	0	2	91	309	597	707	656	595	502	286	26	0
	CDH74	48931	2	44	992	3514	8346	10697	9040	7471	5614	2797	400	14
	CDH80	26682	0	2	301	1746	5099	6872	5035	3677	2723	1180	47	0
Wind	WSAvg	6.6	4.2	6.1	7.4	8.7	9.5	10.0	9.0	7.4	6.0	4.6	3.3	3.2
Precipitation	PrecAvg	41.5	2.4	2.9	2.9	2.0	1.3	2.6	9.4	11.2	4.2	1.0	0.7	1.1
	PrecMax	62.8	5.8	8.2	6.8	8.3	4.5	9.4	24.3	19.6	11.0	3.8	2.8	6.0
	PrecMin	27.9	0.0	0.0	0.0	0.2	0.1	0.3	1.6	3.2	0.6	0.0	0.0	0.0
	PrecStd	8.9	1.8	2.0	1.7	1.7	1.0	2.4	5.2	4.6	2.6	1.0	0.7	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.9	78.9	91.1	101.0	107.6	111.0	105.8	98.7	97.1	94.6	84.1	75.6
		MCWB	53.2	59.1	65.0	68.2	70.7	72.5	78.2	80.8	77.5	73.9	62.9	56.3
	2%	DB	68.8	74.8	87.4	96.9	104.7	107.5	102.0	96.7	95.2	91.6	80.9	72.0
		MCWB	52.3	56.1	63.5	67.1	70.5	73.5	78.7	80.7	77.1	70.7	61.9	54.8
	5%	DB	66.1	71.6	82.8	93.4	101.9	104.4	98.7	95.0	93.4	89.6	78.9	69.7
		MCWB	51.2	54.7	61.6	66.3	70.3	73.5	78.9	79.9	76.4	69.8	60.8	53.7
	10%	DB	62.8	68.5	79.6	89.9	98.8	101.9	96.4	93.1	91.6	87.4	75.6	66.3
		MCWB	50.2	53.9	60.3	65.3	69.5	74.0	78.8	79.3	75.5	68.4	59.0	52.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.3	61.6	68.0	71.1	76.5	81.9	84.0	83.6	82.0	77.0	66.6	59.1
		MCDB	66.2	74.5	86.1	92.4	99.4	99.2	94.9	93.9	91.5	89.7	79.4	71.6
	2%	WB	54.2	58.7	65.3	69.2	74.1	80.2	82.6	82.5	79.9	74.5	64.1	56.8
		MCDB	64.2	69.6	82.3	90.5	97.8	97.6	94.3	92.7	90.8	86.8	77.5	68.2
	5%	WB	52.9	56.8	63.5	67.9	72.6	78.8	81.6	81.5	78.6	72.4	62.2	55.1
		MCDB	62.6	67.2	78.6	88.5	96.0	95.0	92.9	91.2	89.6	84.7	75.1	66.4
	10%	WB	51.2	55.3	61.8	66.6	71.1	77.3	80.6	80.6	77.5	70.2	60.3	53.6
		MCDB	60.7	65.8	75.7	86.1	93.6	93.5	91.4	89.6	87.9	83.6	72.6	63.4
Mean Daily Temperature Range	5% DB	MDBR	22.8	21.3	23.0	24.3	25.1	23.3	17.1	16.0	19.7	24.9	27.0	25.3
		MCDBR	28.2	26.9	28.1	29.3	28.9	27.7	21.0	18.4	21.0	25.9	29.6	29.8
		MCWBR	14.1	12.0	10.7	9.9	9.5	9.2	7.6	7.4	8.2	9.9	13.1	14.8
	5% WB	MCDBR	23.0	22.1	24.2	25.2	25.7	20.6	16.5	16.1	18.1	22.1	26.4	25.8
		MCWBR	12.6	10.8	10.2	9.7	9.6	9.0	7.8	7.7	8.4	9.8	12.6	13.8
Clear-Sky Solar Irradiance	taub	0.426	0.479	0.486	0.509	0.539	0.624	0.696	0.661	0.565	0.520	0.476	0.428	
	taud	1.999	1.915	1.934	1.923	1.868	1.720	1.636	1.703	1.857	1.859	1.901	2.005	
	Ebn at Noon	237	238	250	251	244	223	207	211	225	222	217	227	
	Edh at Noon	45	54	57	60	64	75	81	74	61	56	49	42	
All-Sky Solar Radiation	RadAvg	926	1102	1527	1906	2226	2229	1899	1729	1699	1482	1102	906	
	RadStd	102	152	155	114	108	106	142	134	105	78	62	61	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page